

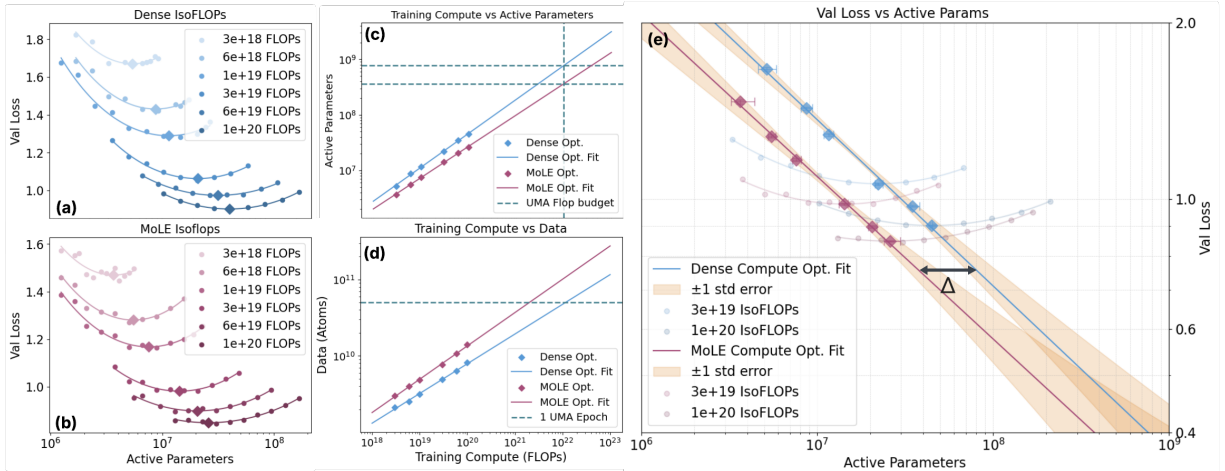


Figure 3 Empirical scaling measurements of dense (blue) vs. MoLE (red) model architectures. FLOPs vs. validation loss for (a) dense and (b) MoLE (8-expert) models. Experiment sets are performed by holding FLOPs constant and varying model size and training data. Diamonds represent the compute optimal frontier. (c) Training compute vs. parameters. Vertical green dotted line represents our estimated training budget of $O(10^{22})$ FLOPs, horizontal dotted lines are corresponding dense and MoLE model sizes. (d) Compute vs. dataset size (atoms) with the green dotted line representing the training FLOPs required for 1 epoch of UMA training data (50B atoms) (e) Overlay of dense vs. MoLE compute optimal frontiers from (a-b) and the fitted power law of validation loss as a function of parameters. A compute optimal MoLE model with $\Delta \approx 2.5\times$ fewer active parameters can achieve an equivalent loss. Fitting details and parameters are described in Appendix C.

Table 2 Test MAE results on held out test splits for materials [59], catalysis [11], molecules [45], molecular crystals [19] and MOFs [68]. All energies are in meV, forces are in meV/Å and stresses are in meV/Å³. Results for UMA are compared against the SOTA literature results. Target accuracies for practical utility are provided as an approximate guide for reference.

Model	Materials						Catalysis				Molecules				Molecular crystals			MOFs	
	WBM Energy/Atom	Forces	Stress	HEA Energy/Atom	Forces	Stress	ID Ads. Energy	Forces	OOD-Both Ads. Energy	Forces	OOD-Comp Total Energy	Forces	PDB-TM Total Energy	Forces	OMC-Test Energy/Atom	Forces	Stress	OOD-L/T Ads. Energy	Forces
UMA																			
UMA-S-1.1	20.2	62.8	4.4	24.9	83.7	3.5	51.5	24.1	68.8	30.7	73.6	8.64	92.7	15.47	1.03	5.04	0.93	289.9	13.3
UMA-S-1.2	17.4	54.2	4.1	17.2	62.8	2.9	47.9	21.6	64.5	27.6	44.2	7.62	78.9	12.69	1.01	3.63	0.84	302.5	15.5
UMA-M-1.1	18.2	50.7	4.2	21.9	69.0	3.5	31.8	15.5	45.5	20.2	59.9	5.44	55.0	10.14	0.84	2.83	0.90	294.1	10.3
Literature																			
eSEN-30M-OMat [23]	16.2	49.6	4.1	20.0	59.5	3.2	-	-	-	-	-	-	-	-	-	-	-	-	-
eqV2-OMat [5]	14.9	46.3	3.6	20.3	47.0	2.7	-	-	-	-	-	-	-	-	-	-	-	-	-
eqV2-OC20 [46]	-	-	-	-	-	-	149.1	11.6	306.5	15.7	-	-	-	-	-	-	-	-	-
GemNet-OC20 [24]	-	-	-	-	-	-	163.5	16.3	343.3	23.1	-	-	-	-	-	-	-	-	-
eqv2-ODAC [68]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	316.0	7.2
eSEN-sm-cons. [45]	-	-	-	-	-	-	-	-	-	-	1.35	7.39	0.83	12.72	-	-	-	-	-
eSEN-S-OMC [19]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.05	5.39	0.94	-	-
Target																			
Practical Utility	10-20	-	-	10-20	-	-	100	-	100	-	43	-	43	-	1-3	-	-	100	-

diverse settings is a non-trivial challenge. Additionally, the datasets differ in size and information content, so we adjust their sampling ratios: 4 for OMat24 and OMol25, 1 for OC20 and ODAC25, and 2 for OMC25. Further details on the datasets and our unified modeling strategy can be found in Appendix B and A respectively.

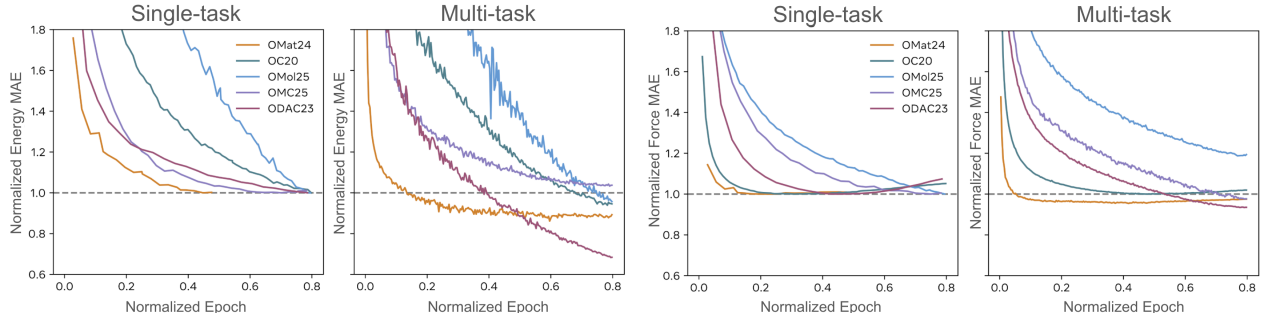


Figure 4 Pre-training curves of UMA-L for both single-task and multi-task models. Errors are normalized based on single-task performance. Note single-task models can overfit (forces on right), and the multi-task model generally converges to lower errors.

4 Results

4.1 Model and Data Scaling

In various domains such as language, empirical power-law relationships have been successfully used [25, 29, 36] to predict the optimal model size and quantity of training data given a fixed compute budget. However, for modeling MLIPs, no models have been trained on the size of UMA’s dataset and very few models have been trained to $>100\text{M}$ parameters [5]; we do not know whether such relationships exist or whether the models will continue to improve with data and parameters. In this study, we show that UMA models do show log-linear scaling behavior as seen in other domains in the FLOP ranges we tested, indicating that greater model capacity is required to fit the UMA dataset. We used these scaling relationships to select the appropriate model size for our dataset and demonstrate the advantage of MoLE over the dense architecture.

Following [25, 29, 36], we first examine the behavior of training compute in FLOPs C as a function of model size N and dataset size D . To determine this, we used total validation loss (computed in the same way as training loss) as the observable quantity. We trained a series of models by sweeping the model parameter size while fixing the training compute budget in FLOPs. This is done by varying the amount of training data for each model, i.e., smaller models receive more training data, while larger models receive less. This is performed for both the dense and MoLE (8 expert) models classes, producing "Iso-FLOPs" curves Figure 3(a,b). The minima of the Iso-FLOPs represent the compute optimal model, or the best achievable loss for the given compute budget (diamonds in Figure 3).

Optimal model size. We fit the Iso-FLOPs minima using power laws [29, 36] to estimate the optimal model and dataset sizes given a fixed compute budget (detailed in Appendix C). Figure 3(c,d) indicates that the dense and MoLE models display convincing log-linear scaling behavior at the scales of our experiments ($10^{18} - 10^{20}$ FLOPs). Given that our training dataset was approximately 50 billion atoms (1 epoch) and our estimated compute budget for pretraining was $O(10^{22})$ FLOPs (extrapolated from our preliminary training runs), Figure 3(c) suggests the largest compute optimal dense model we should train is $\sim 700\text{M}$ parameters corresponding to the UMA-L. In practice, we used the compute optimal model size as a starting point and continued training beyond the compute optimal regime to produce lower losses while minimizing parameter sizes. However, we observed that training on too many epochs can lead to overfitting and significantly deviate from log-linear scaling behavior, prompting us to keep our training within 2-3 epochs.

Dense vs MoLE. If we combine Figures 3(a,b) and fit the Iso-FLOPs minima as a function of loss (Figure 3(e)), we can compare the model sizes needed by the dense and MoLE models to achieve a fixed loss. Effectively, an optimal MoLE model can achieve the same loss as an optimal dense model that is Δ times larger, e.g., $\Delta \approx 2.5 \pm 0.2$ for UMA-M. We observe that this advantage is reduced at larger model sizes; training a 700M active parameter (8 expert - 5.6B parameter total) MoLE model had marginal improvement over the dense version. This effect can be seen in overlaid IsoFLOPs in Figure 3(e). As parameters increase, dense and MoLE performance converge. We hypothesize that we are limited by our dataset’s size, regardless of MoLE or dense architectures.

4.2 Multi-task vs. Single-Task

To explore the benefits of MoLE models for multi-task training, we compare UMA-S (with MoLE) to non-MoLE dense models trained for single/multi-task in Figure 2 (right, top). In this limited parameter regime, multi-task models trained without MoLE always perform worse than single-task (ST) models. With the use of MoLE, UMA-S can achieve results comparable to task specialized small models. We also experimented with varying the number of experts for UMA-S in Figure 2 (right, bottom). A significant improvement in loss is observed when moving from 1 expert to 8. A smaller gain is found with 32 experts, while the gain from 128 experts is negligible. In our experiments UMA-S uses 32 experts.

For large models, multi-task training offers benefits even without MoLE. In Figure 4, we plot the direct-force pre-training curves of UMA-L and single-task models with the same model architecture and size. All metrics are normalized to those achieved with the models trained on single tasks to easily compare their relative performance with multi-task models. We observe that single-task models frequently overfit to forces (OMat overfits upon further training), while the multi-task UMA model does not. Furthermore, UMA achieves lower losses in most cases. The one exception is OMol forces, for which errors are already small (< 10 meV/Å) for both models.

Table 3 Single-GPU simulation speeds for energy-conservative models in *steps per second*: comparing conservative UMA models to the top two models (eSEN and OrbV3) on the Matbench Discovery leaderboard [59] and the MACE materials and molecules models. Benchmarks are run, excluding graph generation, on a single Nvidia H100 80GB GPU using FP32 (TF32-high precision) and torch compile when possible. Test systems are a standard periodic atomic system that have ≈ 50 neighbors per atom with a 6Å cutoff. OOM indicates the model ran out of memory. Refer to Sec.D for more details. Update from March 2026 - UMA-S-1.2 requires code from fairchem-2.16.0+ and includes significant inference time optimizations that are speeds up older model versions. However, we are not updating any previous numbers for good record keeping practices.

Atoms	UMA-S (6.6M)	UMA-S-1.2 (6.6M)	UMA-M (50M)	eSEN-30M- OAM (30M)	Orb-v3 conservative- inf-omat (25M)	MACE- MPA-0 (9M)	MACE- OFF23-L (4.7M)
100	44	53	21	8	77	38	89
1,000	16	24	3	1.7	30	24	20
10,000	1.6	1.3	0.2	OOM	3.7	2.9	OOM
50,000	0.2	0.2	OOM	OOM	OOM	OOM	OOM
100,000	0.1	0.1	OOM	OOM	OOM	OOM	OOM

4.3 Inference Efficiency

The ability to increase the length scale (number of atoms) and time scale of molecular dynamics (MD), relaxations and other types of long running simulations is critical to our design. Leveraging our MoLE strategy, we can pre-merge all the weights and pay no additional penalty in inference time or memory when running a long time-scale simulations on a single system. Here we show that UMA models, despite having very large parameter counts and achieving SOTA accuracy across many domains, are extremely competitive in simulation settings in both speed and raw number of atoms that can fit in memory compared to specialized single-domain models. For example, the UMA-S can simulate 1000 atoms at 16 steps per second (1.4 ns-per-day) and fit system sizes up 100,000 or more atoms in memory on a single 80GB GPU. Lastly, our models are designed to be scaled with multi-GPU parallel inference using the Graph Parallelism strategy [67] described in training; giving the potential to provide orders of magnitude speed-ups when running simulations at the 100k+ atoms scale and unlock new science that is not possible today. We will leave multi-GPU inference outside the scope of this paper and follow-up in a future manuscript.

4.4 Evaluation

Our evaluation contains two main components: (1) a diverse set of held-out test splits (Table 2) and (2) a suite of practically important benchmarks (Table 4). To cover all chemical domains studied: materials, catalysis, molecules, molecular crystals, and MOFs, we propose new test sets and benchmarks, in addition to established benchmarks in existing literature. We highlight selected results below, with more detailed discussion on the results in Appendix E.

Materials. In Table 2, we report the test-set performance on two OOD test sets: WBM [74] and high entropy alloy (HEA). The HEA test set is introduced in this work (Appendix E.2). UMA-M performs on-par with SOTA task specialized models, such as eSEN-30m-OMat. The degraded performance of UMA-L on energy and forces of the HEA dataset suggests overfitting, and its challenging nature. In Table 4, we report benchmark results on the prediction of materials’ thermodynamic stability, thermal conductivity, and vibrational properties. Note for these results, the UMA models are fine-tuned on the MPTrj [17] and sAlex [5, 62] to be consistent with the DFT settings of the public benchmarks. On the popular benchmark Matbench Discovery (MBD) [59], UMA-M achieves the highest F1 score to date. All UMA models of different sizes show strong performance, which suggests that further scaling of model sizes is unlikely to result in better performance on the MBD benchmark. The conserving UMA models (S and M) also excel at phonon-related properties, which are reflected by metrics including κ_{SRME} , free energy MAE, and shear/bulk elasticity MAE.

Catalysis. We evaluate our models on two established catalysis benchmarks: OC20 S2EF [11] and AdsorbML [43]. These benchmarks focus on adsorption energy predictions – that is, the models predict the change in energy as a molecule, known as an adsorbate, comes in contact with a surface. Most previous models on the OC20 benchmark directly predict the adsorption energy given the adsorbate’s position on the surface. Since our model predicts total energy, we make a total energy prediction on the same structure, and another on a clean surface without the adsorbate. These two values are subtracted (along with the energy of the adsorbate in isolation) to predict the adsorption energy. As shown in Table 2, UMA models significantly outperform previous SOTA models on OC20 S2EF adsorption energy prediction – reducing the errors by around 80%. This is not completely unexpected, as total energy models may better account for surface reconstructions than models that directly predict adsorption energy [1]. On the more realistic benchmark of AdsorbML, which evaluates a model’s capability to predict the global minimum adsorption energy, all UMA models outperform the previous SOTA (EquiformerV2). In particular, UMA-L achieves a 25% improvement in the success rate. The strong performance in AdsorbML demonstrates the superior practical utility of UMA for novel catalysis discovery.

Molecules. Initial UMA models were trained on a preview subset ($\approx 70\%$) of the OMol25 dataset, since portions of the dataset were still being calculated when UMA model training began (results in Appendix E). Both UMA-1 and UMA-1.1 were trained on the full OMol25 dataset. In Table 2, we report the prediction performance on two challenging test splits of OMol25: OOD-Comp and OOD PDB-TM. Our results show that UMA-S achieves performance comparable to the domain-specific eSEN-sm-cons, while UMA-M significantly outperforms both models. In Table 4, we evaluate ligand strain and pocket–ligand interaction energy, which are quantities relevant to structure-based drug discovery. In particular, for ligand strain (and, similarly, for conformers in Table 21), UMA-M achieves errors approaching those of the DFT reference, demonstrating its ability to act as a practical surrogate for these calculations. On the distance scaling benchmarks (short and long range) in Table 4, UMA-S slightly surpasses eSEN-sm-cons despite having the same receptive field.

Molecular Crystals. In contrast to materials, molecules, and catalysis, the development and use of universal MLIPs for molecular crystals has been relatively underexplored. In addition to the OMC25 test set, we evaluate UMA’s capability to accurately predict lattice energies, rank, and match structures of the most recent 7th Crystal Structure Prediction (CSP) Blind Test [30]. In all evaluations, UMA-S outperforms the task-specific eSEN-S-OMC baseline, which we trained on the new OMC25 dataset. The high accuracy of lattice energy predictions (≤ 3 kJ/mol) indicates that UMA can be a reliable substitute for DFT in many crystal structure prediction tasks.

Metal Organic Frameworks. Computing the adsorption energy of CO_2 for a MOF sorbent is important for direct-air carbon capture applications and an established benchmark [68]. Similar to OC20, UMA models use total energy predictions to compute the adsorption energy. UMA models perform on par with previous SOTA models on the ID test set (Appendix E) while achieving the best performance on the hardest ODAC OOD test set (Table 2), suggesting improved generalization.

Table 4 Evaluation results on Matbench-Discovery [59], MDR phonon [47], elastic tensor [16, 35], and AdsorbML benchmarks [43]. Results are also provided for a diverse set of molecule [45] and molecular crystal [19, 30] benchmarks. NVE MD [23] tests whether energy conservation is observed when running the model for molecular dynamics. SOTA results from literature are reported where available. Additionally, for the materials evaluations, UMA models were fine-tuned on MPtrj [17] and sAlex [5, 62] to be consistent with the benchmark’s DFT settings.

Model	Materials									Catalysis	Molecules					Molecular Crystals		
	Matbench [59] E_f	RMSD	MAE [eV/atom]	κ_{static} [56]	Phonons [47] ω_{max} [K]	Free Energy [kJ/mol]	Elasticity [16, 35] G_{vph} [GPa]	K_{vph} [GPa]	NVE MD [23] Conserve	AdsorbML [43] Success Rate	OMol25 [45] Ligand-strain [meV]	PDB-pocket [meV]	Dis-SR [meV]	Dis-LR [meV]	NVE MD [23] Conserve	CSP Targets [30] Lattice Energy [kJ/mol]	Kendall Rank	RMSD [Å]
UMA																		
UMA-S-1.1	0.913	0.064	0.020	0.204	18.82	5.48	9.47	5.16	✓	66.80%	4.86	127.7	20.4	194.7	✓	2.13	0.86	0.13
UMA-S-1.2	0.921	0.062	0.019	0.129	10.55	3.24	8.59	4.91	✓	66.19%	4.31	66.6	13.6	32.2	✓	3.02	0.84	0.13
UMA-M-1.1	0.929	0.061	0.018	0.176	14.81	3.87	8.57	4.78	✓	72.25%	2.96	76.8	15.5	138.1	✓	3.24	0.82	0.14
Literature																		
eSEN-30M-OAM [23]	0.925	0.061	0.018	0.170	15.00	4.00	9.13	5.73	✓	-	-	-	-	-	-	-	-	-
ORB v3 [58]	0.905	0.075	0.024	0.210	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SevenNet-MF-ompa [37]	0.901	0.064	0.021	0.317	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GRACE-2L-OAM [8]	0.880	0.067	0.023	0.294	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MACE-MPA-0 [6]	0.852	0.073	0.028	0.412	-	-	-	-	-	-	-	-	-	-	-	-	-	-
eqv2-OC20 [43]	-	-	-	-	-	-	-	-	-	60.80%	-	-	-	-	-	-	-	-
GemNet-OC20 [43]	-	-	-	-	-	-	-	-	-	54.88%	-	-	-	-	-	-	-	-
eSEN-sm-cons. [45]	-	-	-	-	-	-	-	-	-	-	4.66	147.3	21.6	197.0	✓	-	-	-
eSEN-S-OMC [19]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6.18	0.74	0.18

5 Related Work

5.1 Universal MLIPs

The field of MLIPs has been rapidly improving due in part to the availability of larger datasets. A recent example in the materials space is the release of the Alexandria [62] and OMat24 [5] datasets, which quickly led to improved performance on the Matbench-Discovery leaderboard [59]. While MLIPs trained on these materials datasets are often referred to as "universal" because they include nearly all the elements in the periodic table [6, 12, 22, 52], they may not generalize well to other domains such as molecules [38] or surfaces [11]. One reason for this is the chemical, structural, and elemental distribution shifts between materials and other domains such as molecules. Another reason is the level of DFT theory that is considered accurate differs between domains, e.g. PBE for materials and ω B97M-V for molecules. As we demonstrate, training across domains can lead to unified representations and help generalization.

While there have been a number of promising works exploring training MLIPs across multiple domains, it remains an open challenge to demonstrate high-accuracy zero-shot performance. One approach is to pre-train a model using a large corpus of data ($> 100\text{M}$) and fine-tuning for specific tasks [66, 78]. The fine-tuned models are shown to perform significantly better than models trained from scratch. Nevertheless, removing the need for specialization would make these models substantially more useful. Recent studies suggest that achieving zero-shot performance across two DFT tasks may be possible [37, 65].

5.2 Scaling Relations

Empirical scaling laws provide a wealth of insight into the relationships between compute, data, and model size. In LLMs, scaling laws motivated the community to train on more tokens with larger models because performance improvements became predictable [25, 29, 36]. Additionally, scaling relations help practitioners decide on how to best (compute-optimal) allocate resources such as dataset and model size. These types of relations have been explored in MLIPs to compare the asymptotic scaling behavior of different model architectures and training paradigms. For example, [10] showed that equivariant models have different scaling behavior compared to non-equivariant models.